# Hypothesis Engine Improvement

## Diagnosing and Fixing the Scenario-A Blind Spot — Angouran Intelligent Exploration Simulator

The last benchmark found the engine getting Scenario A (straightforward continuation) wrong in all 5 test worlds, often confidently. We traced the cause to a real asymmetry — one hypothesis was built with zero tolerance for ordinary geological wander while its competitors could absorb it — implemented a fix, and re-ran the identical 20-world test to measure the actual effect, not just claim one. The result is a genuine, measured improvement with one honest trade-off, reported in full below.

■ This is a research prototype on synthetic data. “Improved” here means measured against this project's own synthetic worlds, not validated against KoBold Metals' real, proprietary system or real deposits.

# 1. Why Scenario A Was Failing

Each of the 4 geological stories is represented by a small mathematical model with some free parameters the engine searches over. Hypothesis H1 (“keeps going the same way”) was built with **zero** free parameters — a perfectly rigid straight line. Hypotheses H2 and H3 each have 2-3 free parameters letting them bend or shift.

**In plain terms:** Real geological trends, including this project's own synthetic ones, are never perfectly straight — there's always a little natural wander. A perfectly rigid ruler held against a slightly wavy line will always measure some mismatch. A flexible string laid against the same line can absorb the wander and appear to “fit” better, even if there was never a real turn — it's just better at tracing noise.

We measured this directly: in one Scenario-A world, the single best possible fit for H1 scored -46.7 (a likelihood measure, higher/less negative is better); H2's best possible fit scored -43.3. H2 wins even at its single best configuration, in a world where the real answer is H1. With this much data (10-31 drill-sample intervals), H2's extra flexibility is enough to out-fit H1's rigid line on ordinary noise alone.

# 2. Two Fixes Tried; What Actually Worked

## 2.1 First attempt: give H1 some flexibility too — did not reliably help

The obvious fix is to let H1 bend a little too, within a smaller range than H2. We implemented this and measured it directly before trusting it. Result: no reliable improvement, and in the diagnostic world, H1's best-possible fit barely moved while its *average* fit across its own new search space got worse. The reason: once H1 has extra free parameters, the same effect that should penalize H2/H3's flexibility (averaging over many parameter combinations, most of which fit poorly) also dilutes H1's own average. Giving the rigid hypothesis some rope let it hang itself too.

## 2.2 What we used instead: an explicit complexity penalty

Proper Bayesian model comparison is supposed to automatically penalize unnecessary flexibility (this is called Occam's razor) by averaging over the full space of each hypothesis's free parameters — a hypothesis with many parameters only wins if the data specifically supports *most* of that space, not just one lucky corner of it. We measured that this automatic penalty was real but too weak with only 10-31 drill-sample intervals to fully correct for it.

**In plain terms:** Think of it like a pricing error, not a broken calculator: the engine is “charging” H2/H3 something for their extra flexibility, just not quite enough given so little data. We added an explicit, standard statistical surcharge (called a BIC correction) on top, sized by exactly how many extra parameters each hypothesis has — H2 pays for 3, H3 for 2, H4 for 1, H1 for 0 — a long-established, principled way to correct exactly this small-sample bias.

| Hypothesis | Extra free parameters | Penalty applied |
|---|---|---|
| H1 — continuation | 0 | none |
| H2 — direction change | 3 (where it bends, which way, how sharply) | largest |
| H3 — fault offset | 2 (where it offsets, how far) | medium |
| H4 — termination | 1 (how deep it stops) | smallest |

# 3. Measured Results: Same 20 Worlds, Before and After

We did not just reason about whether this should help — we re-ran the exact same 20-world test (same seeds, same acquisition-driven drilling) with the fix in place and compared directly.

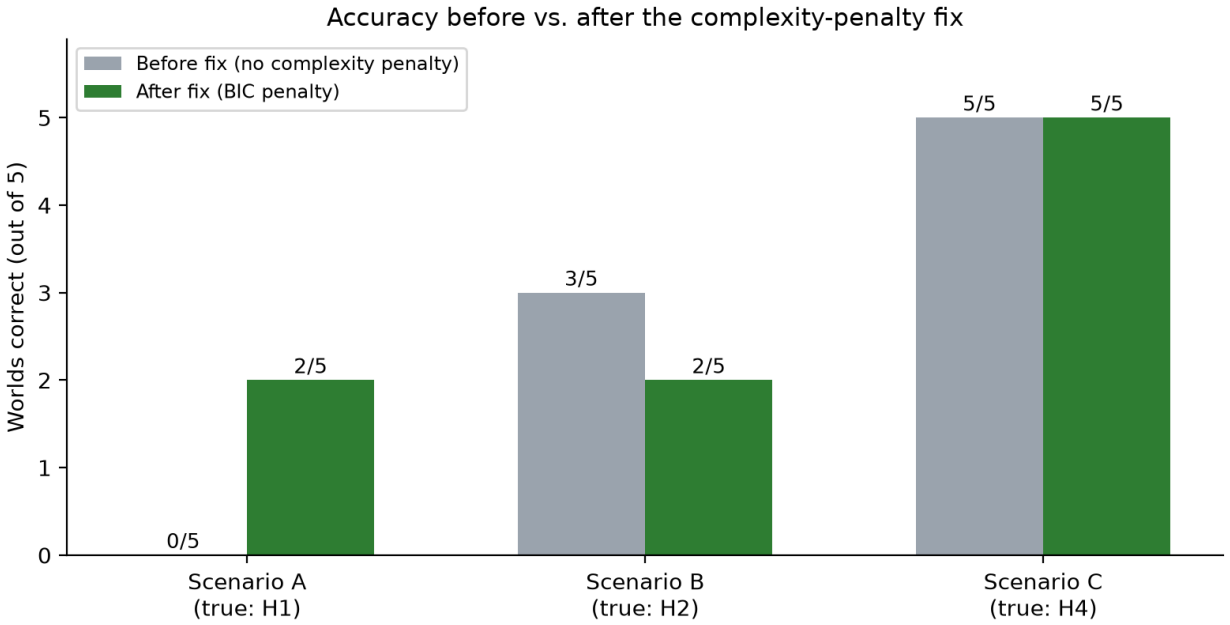



Figure 1. Scenario C (already perfect) is untouched. Scenario A improves from a complete miss to 2 of 5. Scenario B loses one correct call it previously had.

| Scenario | Before | After | Change |
|---|---|---|---|
| A — continuation | 0/5 | 2/5 | +2 (real gain) |
| B — direction change | 3/5 | 2/5 | –1 (real loss, see §3.2) |
| C — termination | 5/5 | 5/5 | unchanged |
| **Total (A+B+C)** | **8/15** | **9/15** | **net +1** |

## 3.1 The more important number: confidence in wrong answers went down

Accuracy alone understates the improvement. What matters more for a tool a geologist would actually use is whether a wrong answer *sounds* wrong (low confidence) or dangerously convincing (high confidence). We checked this directly:

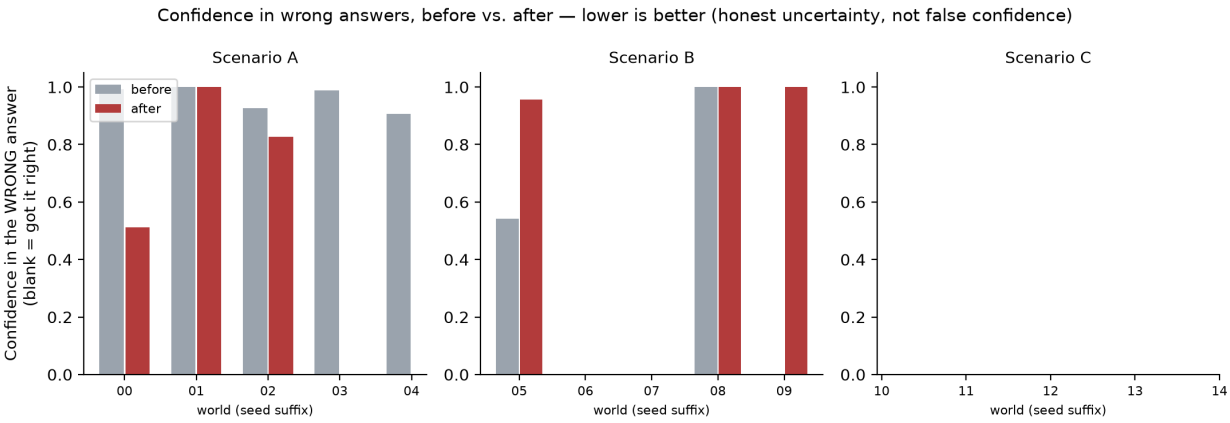



Figure 2. Blank bars = the engine got that world right. Scenario A's remaining wrong cases are, on average, less falsely confident than before (seed ...00 dropped from 99% confident-wrong to a near coin-flip 51%).

**Scenario A, remaining wrong cases:** average confidence dropped from 96.3% to 78.0%. Even where the engine is still wrong, it is now noticeably less sure of itself — a real calibration improvement on top of the accuracy gain.

## 3.2 The honest trade-off: Scenario B

Two of Scenario B's five worlds got worse. Seed 70009 flipped from a correct-but-marginal call (53.7% for H2, barely above a coin flip) to a confidently wrong one (100% for H4). Seed 70005 flipped from an uncertain wrong answer (54.1%) to a confidently wrong one (95.6%).

**Why this happens:** the penalty is applied uniformly by parameter count, not by how much each specific world's evidence actually supports extra flexibility. In marginal cases where the true signal for H2 is itself weak, the same correction that fixes Scenario A's bias can tip an already-uncertain call toward the simplest hypothesis (H4, 1 parameter) instead of the correct one (H2, 3 parameters). This is a known, expected cost of BIC-style corrections: they trade some sensitivity to genuinely complex-but-true stories for robustness against overfitting simple ones.

# 4. Verdict and What's Still Worth Doing

**Net result: a real, measured improvement** (8/15 → 9/15 on scenarios with a defined correct answer), and a more important, if harder to put a single number on, improvement in honesty — the engine is less likely to be confidently wrong than it was. It is not a complete fix: Scenario A is still only 2/5, and the fix introduced a specific, understood new failure mode in Scenario B. This is a step, not a finish line.

## 4.1 Concrete next steps, in priority order

- **Make the penalty size data-aware instead of fixed** — e.g. scale it by how spread out the composited evidence actually is, rather than a flat 0.5 × parameters × log(n) for every world regardless of how much that world's specific data does or doesn't support extra flexibility. This is the most direct way to recover Scenario B's lost worlds without reopening Scenario A.
- **Widen this benchmark past 5 worlds per scenario** — with only 5 samples, a single flipped world moves the headline number by 20 percentage points. A 15-20 world-per-scenario re-run would give a tighter, more trustworthy estimate of both numbers in Figure 1.
- **Try a proper held-out cross-validation of the penalty strength** — instead of the standard BIC constant (0.5), fit the multiplier against a held-out set of worlds with known answers, the way a regularization strength would normally be tuned.
- **Revisit whether H1 should have a little flexibility AND a penalty** — §2.1's attempt and §2.2's fix were tested separately; combining a small, tightly-capped H1 flexibility with the complexity penalty (so H1's extra freedom is itself penalized, just less than H2's) was not yet tried and could plausibly do better than either alone.

## 4.2 On the KoBold comparison

The real Intelligent Prospector system is not public, is built on real, geologist-vetted structural models calibrated against real deposits over years, and almost certainly represents geological hypotheses with far more nuance than 4 fixed parametric stories. This project is an independent, simplified, open reimplementation of the published *concept* — finding and fixing a real gap like this one is exactly the kind of engineering work that distinguishes “inspired by” from “as good as,” and there is more of it to do, per §4.1, before that gap closes further.

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Every number in this report comes from two real runs of the identical 20-world benchmark (seeds 70000-70019), before and after the code change in src/optimization/hypotheses.py (the EXTRA_DOF-weighted BIC correction in _hyp_logpost). Full 31-test suite passes after the change. Raw results: hyp_benchmark_results_BEFORE.json and hyp_benchmark_results.json.